

From Data Cleaning to Animal-ID Matching

A leakage-aware evaluation of what milking-session records can—and cannot—tell us about an individual cow.

Main finding: consistent data and better matching features improved retrieval, but the strongest ID result is still a pilot—not reliable automatic assignment.

Analysis notebook · Fall 2026

8.50M
raw milking-session rows

60,653
exact duplicate rows removed

20.02%
raw records missing AnimalId

9,087
known cow IDs before splitting

1 · Clean and audit

8,495,421 × 11 The CSV contains animal/lactation metadata, event date, flow, yield, duration, and milking number.

8,495,421 raw rows — 60,653 exact full-row duplicates = 8,434,768 rows retained

- Preserved AnimalId as text** so large identifier values were not treated as measurements.
- Standardized types:** coerced numeric fields to numeric, parsed EventDate as a date, and trimmed ReproductionStatus; blank text was marked missing.
- Removed exact duplicates only:** rows had to match in every column; repeated AnimalId values were expected and retained.
- Checked impossible values:** nonpositive lactation/milking numbers and negative days, flows, yields, or duration. The audit found **0 violations**, so 0 rows were removed by these rules.
- Reviewed duplicate-like session keys** (same cow/date/milking) but kept them for review rather than deleting records on a partial key.

Outlier and missing-data policy: the recorded cleaning run does not show statistical outlier trimming or imputation. The 1,696,870 rows with missing AnimalId were retained separately; an unknown cow ID is not a zero or a safe value to invent.

2 · What was missing?

Missingness was concentrated in cow/lactation metadata. Most session measurements and all event dates were populated in the raw-file audit.

RAW FIELD	MISSING ROWS	SHARE
AnimalId	1,701,003	20.02%
LactationNumber	1,701,003	20.02%
DaysInMilk	1,701,007	20.02%
ReproductionStatus	1,701,003	20.02%
Avgmilkflow	172	<0.01%
EventDate / other session measures	0	0%

Counts above are from the raw 8.50M-row audit; after exact deduplication, 1,696,870 rows remained without AnimalId.

3 · Split without cow leakage

All sessions from one known cow stayed together in the primary train/test partition. This prevents the same cow from appearing in both partitions when testing behavior-group generalization.

80% · 7,270 cows
training IDs · 5,396,948 rows

20% · 1,817 cows
held-out IDs · 1,340,950 rows

- Split unique known AnimalId values with **random_state=42**.
- Confirmed **0 shared cow IDs** between train and test.
- Kept the **1,696,870 missing-ID rows** as a separate inference pool, not as training labels.

A second split served the matching task: within known cows, session keys were deterministically hashed and roughly 1-in-5 sessions held out as queries; the other sessions formed the cow profile/gallery. Gallery and query sessions were distinct.

Success · Broad behavior groups

This is a **3-class behavior classification** task, not exact cow identification. Three cow groups were described using mean milk flow, session yield, and duration; KNN used those session measurements.

k = 9 selected 66.51% test accuracy 62.72% balanced accuracy

The held-out majority-class baseline was **43.38%**. The KNN improved on it; class-level F1 ranged from 0.47 to 0.74, showing the groups were not equally easy to recognize.

k was selected by inner validation (tested 1, 3, 5, 9); k=9 led with 68% accuracy / 64% balanced accuracy there. Final held-out evaluation used 8,907 sampled sessions from 1,807 cows.

What the split lets us claim

The cow-level holdout supports evaluation of broad behavior groups on cows not used to fit group centers. The individual-ID retrieval tests answer a different question: can a session match one of the cows that already has a labeled history?

Neither task validates the true IDs of the missing-ID records, because those labels are unavailable for scoring.

4 · Predicting an exact AnimalId

Nearest-neighbor retrieval compares a query session with known cow profiles/sessions. Exact accuracy means the top-ranked candidate ID is correct; top-five means the true cow appears among five distinct candidates.

8 NUMERIC SESSION FEATURES lactation, days in milk, two flow measures, early and total yield, duration, and milking number.

- Standardized measurements so large-unit features would not dominate distance.
- Tested Manhattan/Euclidean-style distances and feature subsets.
- Added timing, reproductive status, and four derived production-shape features.
- Expanded the gallery from 5–20 to as many as 100 sessions per cow.

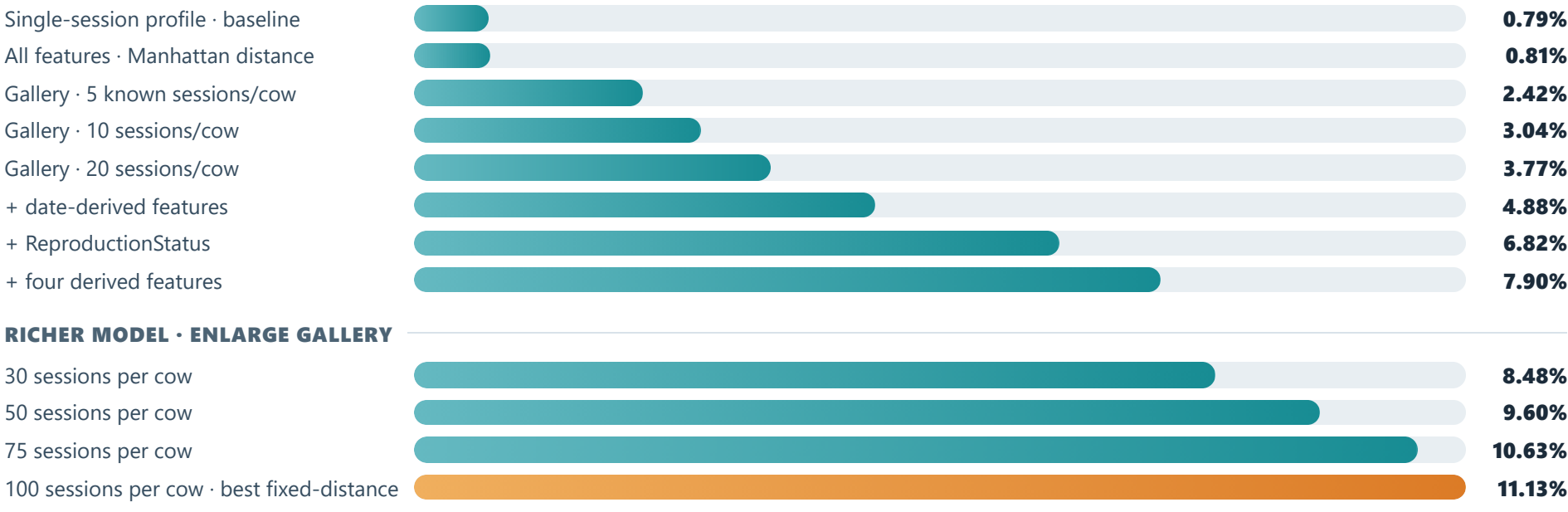
Weak attempts · What did not help

- Simple profile matching:** 0.79% exact accuracy; top-five coverage 2.11%.
- Changing the distance alone:** all-feature Manhattan reached only 0.81%; production-only/context subsets were about 0.18–0.71%.
- Five-session majority vote:** 0.46% exact accuracy, below the 0.79% single-session baseline. Repeated guesses did not reliably fix the match.
- Date alone:** 0.014% exact accuracy and 0.13% top-five; timing needs measurement context.
- NCA sweep:** 27 configurations tested. Tuning did not beat the simple NCA result materially; the best tuned run was 22.86% exact.

5 · Accuracy experiments: useful gains, important limits

Exact ID retrieval on the larger query sample

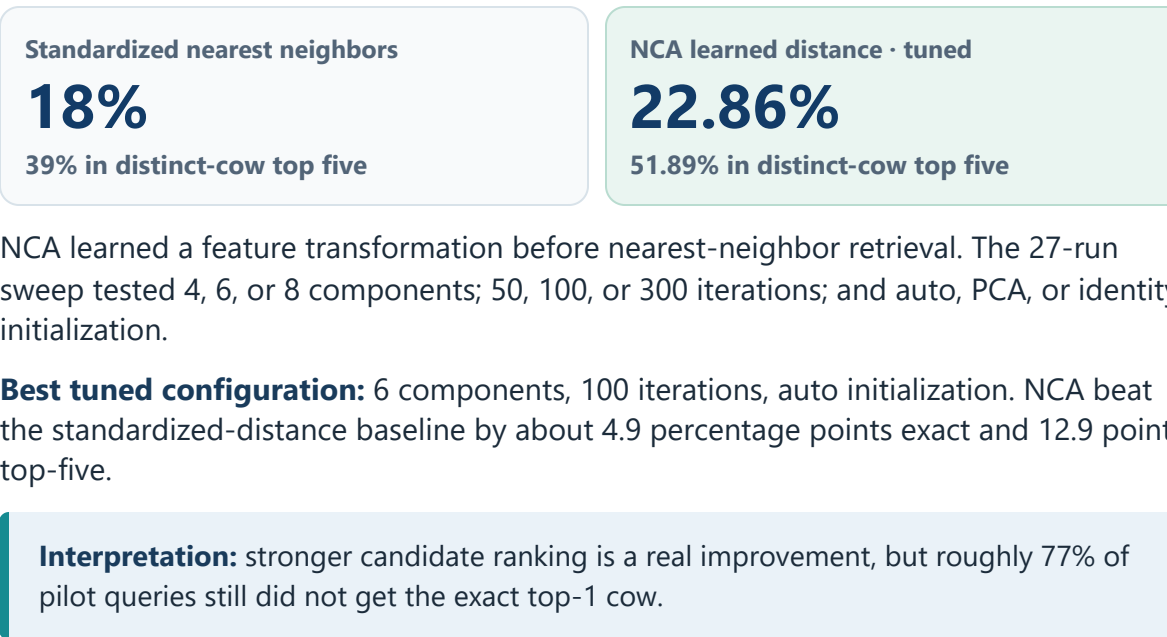
The sequence below tracks the notebook's fixed-distance experiments on the same query sample (35,582 sessions; up to five query sessions per cow). Gallery and feature additions are cumulative where shown.



Date plus measurements raised exact accuracy 3.81% → 4.88%; date alone was ineffective. Adding reproductive status and derived features raised it further to 7.90%. A larger gallery then helped incrementally.

Controlled NCA pilot

Same 2,463 held-out sessions and same 499-cow candidate set for both models. This is a smaller pilot than the larger-sample progression at left.



Keep the outcomes separate

- Worked: broad behavior classification**
66.51% accuracy vs 43.38% majority baseline; predicts one of three behavior groups, not a cow's unique ID.
- Helped: more session evidence**
With richer features, growing the cow gallery to 100 sessions raised exact retrieval to 11.13%.
- Helped: learned distance (pilot)**
NCA increased exact and distinct-cow top-five performance over standardized nearest neighbors on the same pilot queries.
- Did not solve: unique ID from one record**
Feature/distance tweaks alone barely moved the larger-sample baseline; majority voting and date-only matching performed poorly.
- Still unknown: missing-ID truth**
The 20.02% unassigned records cannot be validated without their true IDs. A cow with no known labeled gallery session is not identifiable by this approach.

Bottom line

Cleaning removed exact duplicates and made types consistent without fabricating missing IDs or discarding valid-looking sessions. Leakage-aware splits made the behavior-group result credible. For exact cow assignment, more labeled sessions and NCA improved ranking—but accuracy is not yet sufficient to automatically fill missing AnimalId values.

22.86%
best tuned NCA exact match
pilot: 2,463 queries · 499 cows